

Question 1

If $u = \tan^{-1} \left[\frac{x^3 + y^3}{x - y} \right]$, prove that:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \sin u \cos 3u$$

Complete Step-by-Step Derivation

Step 1: Transform the Function to Remove the Non-Linear Wrapper

The function u is not directly homogeneous because of the outer inverse tangent function. We eliminate this wrapper by taking the tangent of both sides to isolate the core algebraic expression into a new substitute function z :

$$z = \tan u = \frac{x^3 + y^3}{x - y}$$

Step 2: Determine the Degree of Homogeneity (n)

Let the algebraic function be represented as $f(x, y)$:

$$f(x, y) = \frac{x^3 + y^3}{x - y}$$

To find the degree of homogeneity, substitute $x \rightarrow xt$ and $y \rightarrow yt$:

$$f(xt, yt) = \frac{(xt)^3 + (yt)^3}{(xt) - (yt)}$$

Factor out the scaling parameter t from both the numerator and the denominator:

$$f(xt, yt) = \frac{t^3(x^3 + y^3)}{t^1(x - y)}$$

Simplify the powers of t :

$$f(xt, yt) = t^{3-1} \left[\frac{x^3 + y^3}{x - y} \right]$$

$$f(xt, yt) = t^2 f(x, y)$$

Since t factors out with a power of 2, the substitute function z is a homogeneous function with a degree of $n = 2$.

Step 3: Compute the First-Order Value Expression $g(u)$

According to the Modified Euler's Theorem, when a non-homogeneous function is transformed via $z = f(u)$, the first-order differential expression equates to a single-variable expression $g(u)$:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)} = g(u)$$

From our setup:

$$f(u) = \tan u$$

Differentiating $f(u)$ with respect to u gives:

$$f'(u) = \sec^2 u$$

Substitute $n = 2$, $f(u)$, and $f'(u)$ into the formula to evaluate $g(u)$:

$$g(u) = 2 \cdot \frac{\tan u}{\sec^2 u}$$

Convert the trigonometric terms to standard sine and cosine identities to simplify the fraction:

$$\begin{aligned} g(u) &= 2 \cdot \frac{\left(\frac{\sin u}{\cos u}\right)}{\left(\frac{1}{\cos^2 u}\right)} \\ g(u) &= 2 \cdot \frac{\sin u}{\cos u} \cdot \cos^2 u \\ g(u) &= 2 \sin u \cos u \end{aligned}$$

Apply the double-angle trigonometric identity ($2 \sin u \cos u = \sin 2u$):

$$g(u) = \sin 2u$$

Step 4: Apply the Second-Order Modified Euler's Corollary

When a composite multi-variable function produces a first-order expression equal to $g(u)$, its corresponding second-order operator expansion simplifies to the following identity:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) \cdot [g'(u) - 1]$$

Find the derivative of our calculated $g(u)$ with respect to u using the chain rule:

$$g'(u) = \frac{d}{du}(\sin 2u) = 2 \cos 2u$$

Substitute the expressions for $g(u)$ and $g'(u)$ into the right-hand side (RHS) of the second-order corollary:

$$\text{RHS} = \sin 2u \cdot (2 \cos 2u - 1)$$

Step 5: Expand and Simplify to Match the Target Identity

Multiply the terms across the bracket:

$$\text{RHS} = 2 \sin 2u \cos 2u - \sin 2u$$

Apply the standard sine double-angle identity to the first term ($2 \sin 2u \cos 2u = \sin 4u$):

$$\text{RHS} = \sin 4u - \sin 2u$$

Apply the difference-of-sines identity, $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$, where $C = 4u$ and $D = 2u$:

$$\text{RHS} = 2 \cos\left(\frac{4u+2u}{2}\right) \sin\left(\frac{4u-2u}{2}\right)$$

$$\text{RHS} = 2 \cos\left(\frac{6u}{2}\right) \sin\left(\frac{2u}{2}\right)$$

$$\text{RHS} = 2 \cos 3u \sin u$$

Rearrange the terms to match the required problem structure:

$$\text{RHS} = 2 \sin u \cos 3u$$

Equating this final expression back to the differential operator statement completes our proof:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \sin u \cos 3u$$

Hence, proved.